

Assignment 3 Report

Deep Learning for Computer Vision (LeNet-5 and YOLOv1-style Detection)

This report presents the implementation, training, evaluation, and comparison of two influential convolutional neural network architectures required for the Deep Learning for Computer Vision assignment: LeNet-5 and YOLOv1-style object detection. The objective of this assignment was to understand the evolution of CNN architectures, analyze their architectural innovations, and evaluate their performance on realistic datasets using PyTorch implementations.

1. Assignment Objectives

The objectives of this assignment were:

- Explain the motivation behind CNN architectures.
- Describe the architectural innovations introduced by the selected models.
- Implement deep learning models using PyTorch.
- Train and evaluate the models using realistic datasets.
- Analyze strengths, weaknesses, and computational tradeoffs.
- Present findings professionally.

2. Selected Architectures

The architectures selected for this assignment were:

LeNet-5 (1998) – one of the earliest successful convolutional neural networks for handwritten digit recognition.

YOLOv1-style Detection (2016) – a real-time object detection framework that introduced single-stage object detection using one forward pass through a neural network.

3. Background and Motivation of LeNet-5

Before LeNet-style convolutional neural networks, computer vision systems for handwritten recognition mainly depended on handcrafted features such as edge detectors, shape templates, and manually designed descriptors. These systems required significant feature engineering and often lacked generalization ability.

LeNet-5, proposed by Yann LeCun in 1998, demonstrated that convolutional neural networks could automatically learn hierarchical visual features directly from raw image pixels. The architecture introduced several important concepts including local receptive fields, shared weights, and pooling/subsampling operations.

The architecture followed a simple pattern:

Conv → Pool → Conv → Pool → Fully Connected → Output

In this project, the implementation was adapted for Fashion-MNIST datasets with 28×28 grayscale images while preserving the original architecture topology.

4. Architectural Innovations of LeNet-5

LeNet-5 introduced several important innovations:

- Convolutional feature extraction reduced parameter count compared to fully connected networks.
- Weight sharing allowed translation-equivariant feature detection.
- Pooling/subsampling reduced spatial dimensions and improved translation tolerance.
- End-to-end training enabled automatic feature learning directly from image data.

5. Background and Motivation of YOLOv1

Before YOLO, many object detection systems used two-stage pipelines. Models such as R-CNN, Fast R-CNN, and Faster R-CNN first generated region proposals and then classified each region separately. While these approaches achieved high accuracy, they suffered from high latency because multiple computations were required for each image.

YOLO (You Only Look Once), introduced in 2016, reframed object detection as a single regression problem. Instead of generating proposals first, YOLO predicts bounding boxes and class probabilities directly from one forward pass through the network.

The input image is divided into an $S \times S$ grid. Each grid cell predicts:

- Bounding box coordinates
- Objectness/confidence score
- Class probabilities

This design made YOLO significantly faster and suitable for real-time applications.

6. Architectural Innovations of YOLOv1

YOLOv1 introduced several major innovations:

- Single-stage object detection using one forward pass.
- Unified loss function combining localization, objectness, and classification loss.
- Global image reasoning through dense prediction.
- End-to-end trainable detection pipeline without external proposal generation.

In this assignment, the implementation used a simplified TinyYOLOv1-style architecture with one predicted box per grid cell to reduce complexity and improve readability for coursework purposes.

7. Experimental Setup

Two different datasets and tasks were used:

Component	LeNet-5	YOLOv1-style
Dataset	Fashion-MNIST	PASCAL VOC 2007
Task	Image Classification	Object Detection
Input Type	28×28 grayscale images	Natural scene images
Primary Metric	Top-1 Accuracy	mAP@0.5
Implementation	models/lenet.py	TinyYoloV1 + YoloV1Loss

8. Experimental Results

LeNet-5 Results

The LeNet-5 implementation achieved strong classification performance on Fashion-MNIST. The best test accuracy obtained was 90.60% at epoch 13. Training remained stable throughout the experiment, and only mild overfitting was observed in later epochs.

YOLOv1-style Results

The TinyYOLOv1-style detector achieved a best validation loss of 3.338 at epoch 16. However, the mean

Average Precision (mAP@0.5) on the VOC 2007 validation dataset was approximately 0.35%, which is very low compared to production-level object detectors.

This low mAP was expected because the implementation used:

- A simplified single-scale detection framework
- One box prediction per grid cell ($B=1$)
- Limited training epochs and reduced complexity for coursework

Aspect	LeNet-5	YOLOv1-style
Main Output	Single class label	Bounding boxes + classes
Task Complexity	Lower	Higher
Computation	Lightweight	More computationally expensive
Observed Performance	~90.6% accuracy	~0.35% mAP@0.5
Main Strength	Efficient image classification	Real-time detection

9. Strengths, Weaknesses, and Tradeoffs

Strengths of LeNet-5

- Simple and computationally efficient architecture.
- Strong inductive bias for image recognition tasks.
- Fast inference and stable training behavior.

Limitations of LeNet-5

- Limited capacity for complex large-scale datasets.
- Shallow depth compared to modern CNNs.

Strengths of YOLOv1

- Real-time object detection capability.
- Unified end-to-end detection framework.
- Lower latency compared to two-stage detectors.

Limitations of YOLOv1

- Poor small-object localization in early versions.
- Grid-cell limitations when multiple objects fall into the same cell.
- Lower accuracy compared to modern detectors.

10. Conclusion

This assignment demonstrated the evolution of convolutional neural networks from early image classification architectures to modern real-time object detectors.

LeNet-5 showed how convolution, pooling, and weight sharing can efficiently learn visual features from image data and achieve strong classification performance on Fashion-MNIST.

YOLOv1-style detection demonstrated the advantages and challenges of single-stage object detection. While the simplified implementation achieved low detection accuracy, it successfully illustrated the core motivation of YOLO: achieving real-time object detection by removing the proposal generation stage.

Overall, the experiments highlighted the importance of architectural design choices, dataset complexity, and computational tradeoffs in deep learning for computer vision.

11. References

- LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-Based Learning Applied to Document Recognition.
- Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection.
- Fashion-MNIST Dataset.
- PASCAL VOC 2007 Dataset.